



The cost of age extension

Even if medical research manages to extend human life, the costs could outweigh the benefits. <https://digit.in/mar20-57>



Escaping the doom of man

If we actually cure ageing, will we still remain human any more? Will we, like Dorian Gray, lose our souls? <https://digit.in/mar20-58>



SENESCENCE

Both an organism and the individual cells begin to deteriorate with an increase in age. Researchers are looking into the exact mechanisms, and if getting old biologically is just another disease that can be cured.

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rom 30 years in the neolithic period, the global average life expectancy has today reached a little more than 72 years. This means that for millions of years, humans have been reproducing when they were young and fit, and the process of natural selection did not really have the chance of weeding out the genes that cause problems later in life. Such a theory for explaining why humans degenerate with age was first put forward by Peter Medawar in 1952, and forms the basis for most modern theories on ageing. After humans reach peak fitness, and beyond the age of reproduction, the

body begins to rapidly deteriorate. The quality of life for the elderly is not great. They may experience neurological deterioration, an increased susceptibility to heart disease, and are more likely to get cancer. There are a number of age-related diseases including osteoporosis, alzheimer's and cataracts, with an exponential increase in incidence alongside ageing. Around the world, a lot of resources are spent in taking care of the elderly. What if each of these diseases are a symptom, while the actual condition that needs to be tackled is ageing? The basic idea is that if we figure out and treat the underlying mechanism that causes ageing, then maybe these resources to care for the elderly can be applied elsewhere. However, this may be a flawed line of thinking. If ageing is

a disease, it is something that affects all human beings, as well as all multicellular organisms. Those who remain healthy and active in their old age are considered to be blessed with good genes, but they show signs of biological ageing as well, such as wrinkles. Now even an immortal organism (the hydra may just be one such), experiences ageing. Medawar elegantly explained the reason to coin a term to describe the deterioration associated with ageing, "we obviously need a word for mere ageing, and I propose to use 'ageing' itself for just that purpose. 'Ageing' hereafter stands for mere ageing, and has no other innuendo. I shall use the word 'senescence' to mean ageing accompanied by that decline of bodily faculties and sensibilities and energies which ageing colloquially entails." Gerontologists, or the scientists that specialise in the study of senescence, are not interested in discovering the elixir of life. Instead, they are more interested in coming up with treatments that can extend the average life span from